

MTH401 FINAL TERM SOLVED MCQs

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The annihilator operator of the function $y = e^{-6x}$ is

$(D - 6)^2$

$(D - 6)^3$

$D - 6$

Correct answer

$D + 6$

After converting the given differential equation $4y'' + 64y = \sec 3x$ into standard form, the function $f(x)$ is

$(\sec 3x)/4$

Correct answer

$(\sec 3x)/64$

None of them

$\sec x$

Wronskian of the function $y_c = c_1 + c_2 \cos x + c_3 \sin x$ is

0

1

Correct answer

2

3

In the infinite series of $(x-a)$ which can be written as

$$\sum_{n=0}^{\infty} c_n (x-a)^n = c_0 + c_1 (x-a) + c_2 (x-a)^2 + \dots$$

the number **a** is called the

Radius of power series

Centre of power series

Base of power series

None of them

Correct answer

$$\sum_{n=1}^{\infty} a_n (x-a)^n$$

Suppose that a power series is represented by a function “f” whose domain is the interval of the convergence of the power series. That function “f” is continuous, differentiable and integrable on

(a + R, a - R)

(R - a, R + a)

(a - R, a + R)

Correct answer

None of them

The interval of convergence for the function **secx** is

$(-\pi, +\pi)$

$(-\frac{\pi}{2}, \frac{\pi}{2})$

correct answer

$(\frac{\pi}{2}, \pi)$

None of them

$$\frac{dy}{dx} - 2xy = 0$$

The solution of the linear first order differential equation is

$$y = e^{x^2}$$

Correct answer

$$y = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

Both $y = e^{x^2}$ & $y = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$
 None of them

$$X = L_{\gamma} - \frac{1}{C_{\gamma}}$$

The quantity is called
Reactance of circuit Correct answer
 Impedance of circuit
 Quasi of circuit
 None of them

$$\frac{d^2x}{dt^2} + 2\lambda \frac{dx}{dt} + \omega^2 x = 0$$

For the equation of free damped motion the roots are
 $m_1 = -\lambda + \sqrt{\lambda^2 - \omega^2}$ & $m_2 = -\lambda - \sqrt{\lambda^2 - \omega^2}$ If $\lambda^2 - \omega^2 > 0$ then system is said to be

Under damped
Over damped Correct answer
 Critically damped
 None of them

The time interval between two successive maxima of $x(t) = Ae^{-\lambda t} \sin[\sqrt{\omega^2 - \lambda^2} t + \phi]$ is called
Phase period Correct answer
 Quasi-period
 Both the period
 None of them

$$\frac{d^2x}{dt^2} + 5 \frac{dx}{dt} + 4x = 0$$

The given differential equation is
Over damped

Critically damped
Under damped
None of them

The standard unit for measurement of inductance is
Volt
Ohms
Henry Correct answer
None of them

Which of the rule in matrices under multiplication does not hold true?
Commutative law
Associative law
Identity law
None of them

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \end{bmatrix} \& B = \begin{bmatrix} x & y & z & a \\ p & q & r & b \\ l & mn & o \end{bmatrix}$$

If then the order of $matrix\ A \times B$ is

2×4
Correct answer
 2×3
 3×3
None of them

$$\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 & -7 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} 4 \\ 8 \end{pmatrix} \sin t$$

The given system without the use of matrices is
 $\frac{dx}{dt} = 3x - 7y + 4\sin 2t$; $\frac{dy}{dt} = x + y + 8\cos 2t$

$$\frac{dx}{dt} = 3x - 7y + 4\sin t; \quad \frac{dy}{dt} = x + y + 8\cos t$$

$$\frac{dx}{dt} = 3x - 7y + 4\sin t; \quad \frac{dy}{dt} = x + y + 8\sin t$$

Correct answer

None of them

Suppose that $\{X_1, X_2, X_3, \dots, X_n\}$ is a set of n solutions vectors on an interval I , of a homogeneous system $X' = AX$. The set is said to be a fundamental set of solutions of the system on the interval I if the solution vectors are

Linearly dependent

Linearly independent

Correct answer

Homogeneous

None of them

The coefficient matrix of the following homogeneous system of differential equation

$$\frac{dx}{dt} = 3x + 2y, \quad \frac{dy}{dt} = x + 2y$$

$$\begin{bmatrix} 3 & 2 \\ 2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 2 \\ 1 & 2 \end{bmatrix}$$

Correct answer

None of them

$$A = \begin{bmatrix} 3 & -18 \\ 2 & -9 \end{bmatrix}$$

The matrix has an eigen value of multiplicity

1

2

3

4

$$A = \begin{bmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & -2 & 1 \end{bmatrix}$$

The matrix
Single root of A
triple root of A
double root of A
None of them

has eigen values $\lambda = -1, -1, 5$ where $\lambda = -1$ is a

$$\begin{vmatrix} 4-\lambda & 1 & 0 \\ 0 & 4-\lambda & 1 \\ 0 & 0 & 4-\lambda \end{vmatrix} = 0 \text{ gives}$$

$\lambda = 4$ of multiplicity of 1

$\lambda = 4$ of multiplicity of 2

$\lambda = 4$ of multiplicity of 3

None of them

$$\frac{dy}{dt} = 2x, \frac{dx}{dt} = 3y$$

For the system of differential equations

the independent variable(s) is (are)

x, t

y, t

x, y

t

$$\frac{dy}{dt} = 2x, \frac{dx}{dt} = 3y$$

For the system of differential equations

the dependent variable(s) is (are)

x, t

y, t

x, y

t

If $\mathbf{L}$ denote the linear differential operators with constant coefficients, then $L_1L_4 - L_2L_3$ represents the

$$\begin{vmatrix} L_1 & L_2 \\ L_4 & L_3 \end{vmatrix}$$

$$\begin{vmatrix} L_1 & L_3 \\ L_4 & L_2 \end{vmatrix}$$

$$\begin{vmatrix} L_1 & L_2 \\ L_3 & L_4 \end{vmatrix}$$

Correct answer

None of them

Wronskian of x, x^2 is

x

x^2

x^3

0

$$\frac{dy}{dx} = \frac{x+y}{x}$$

The general solution of differential equation .is given by

$$e^{\frac{y}{x}} = cx$$

$$e^{\frac{y}{x}} = cy$$

$$e^{\frac{x}{y}} = cx$$

$$e^{-\frac{x}{y}} = cx$$

The form of the exact solution to

$$2 \frac{dy}{dx} + 3y = e^{-x}, y(0) = 5$$

is

$$Ae^{-1.5x} + Bxe^{-x}$$

$$Ae^{1.5x} + Be^{-x}$$

$$Ae^{1.5x} + Bxe^{-x}$$

$$Ae^{-1.5x} + Be^{-x}$$

If m and n are non negative integers and $P_n(x)$ is a Legendre's polynomial then

$$\int_{-1}^1 P_m(x)P_n(x)dx = 0 \quad \text{for } m \neq n$$

Correct answer

$$\int_{-1}^1 P_m(x)P_n(x)dx = 0 \quad \text{for } m = n$$

$$\int_{-1}^1 P_m(x)P_n(x)dx = 0 \quad \text{for } m < 0$$

$$\int_{-1}^1 P_m(x)P_n(x)dx = 0 \quad \text{for } n < 0$$

If A is a square matrix and its determinant is zero, then

A is singular matrix. **Correct answer**

A is non singular matrix.

A is scalar matrix.

A is diagonal matrix.

An electronic component of an electronic circuit that has the ability to store charge and opposes any change of voltage in the circuit is called

Inductor

Resistor

Capacitor correct answer

None of them

Operator method is the method of the solution of a system of linear homogeneous or linear non-homogeneous differential equations which is based on the process of systematic elimination of the

Dependent variables correct answer

Independent variable

Choice variable

None of them

Any linear differential equation of the form

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1} x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 x \frac{dy}{dx} + a_0 y = g(x) \text{ where } a_0, a_1, a_2, \dots, a_n \text{ are constants.}$$

is called

Homogeneous equation

Polar equation

Equi-dimensionl equation or Cauchy eular

correct answer

None of them

$$A = \begin{pmatrix} 3 & 4 \\ - & \end{pmatrix}$$

For eigen values $\lambda = 5, 5$ of a matirx ,there exists eigen vectors.

infinite

one

two

three

Ordinary points of $(x^2 - 64)(x^2 - 36)y' + xy' - y = 0$ are

0,1

8,-8

6,-6

None of others.

A singular point $x = x_0$ of the given equation $a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$ is said to be a regular singular point if

$(x - x_0)P(x)$ is analytic at x_0

$(x - x_0)Q(x)$ is analytic at x_0

$(x - x_0)P(x)$ & $(x - x_0)^2Q(x)$ are analytic at x_0 . Correct answer

None of them

Singular points of the equation $(x^2 - 4)^2 y'' + (x - 2)y' + y = 0$ are

$x = -2, 2$

None of them

$x = 2$

$x = -2$

$$A = \begin{bmatrix} 1 & 0 \\ 0 & \end{bmatrix}$$

The matrix has

Real and unequal value

Repeated & real eigen value

Complex eigen value

None of them

Let λ be an eigen value of a non zero square matrix A. Then the equation $\det(A - \lambda I) = 0$ is called

Trivial equation

Characteristics equation

Non-trivial equation correct answer

None of them

If $y_1 = x^2$ is solution of the differential equation, then formula for finding Second solution of $x^2 y'' - 2y = 0$ is

$$y_2 = x^2 \int \frac{e^{-2x}}{x^2} dx$$

$$y_2 = x \int \frac{e^{-2x}}{x^2} dx$$

$$y_2 = x^2 \int \frac{e^x}{x^4} dx$$

$$y_2 = x^2 \int \frac{2}{x^4} dx$$

For $y \sin^2 x - y^2 \cos x = c$ where $y(0) = 3$; the value of c is -----.

9

-9
10
-10

A differential equation $M(x, y) dx + N(x, y) dy = 0$ is said to be an exact if -----.

$$\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \quad \text{Correct answer}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$$

Logistics equation is an Example of

Linear

Non linear

Bernoullis

One more name was there I forgot it.

$$\frac{dy}{dx} + (\cot x) y = \cos^2(x)$$

The integrating factor of the differential equation is -----

$$\mu = \ln |\sin x|$$

$$\mu = \ln |\cos x|$$

$$\mu = \sin x$$

$$\mu = \cos x$$

2 Marks

Give two examples of Bessel's Differential Equation?
What is wronskian?

Give principle of superposition to find out any homogeneous equation

Define general linear equation of order n?

Marks 3

$$\frac{d^3 y}{dx^3} - 4 \frac{d^2 y}{dx^2} + 5 \frac{dy}{dx} = 4x$$

Write the differential equation

differential operator with constant coefficients

in the form $L(y) = g(x)$ where L is a

Marks 5

Deduce the Special Case of Logistic Equation “Epidemic spread”?

Write down the system of differential equations

$$\frac{dx}{dt} = 6x + y, \quad \frac{dy}{dt} = x + 3y - 9t - 9$$

in form of $X' = AX + F(t)$

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